# IPL Batting Dataset Description

**The dataset contains batting statistics of IPL (Indian Premier League) players from multiple seasons (2008–2025). Each row represents the batting performance of a player in a specific IPL season for a particular team.**

**The dataset is designed for:**

- **Descriptive Data Analysis**
- **Sports Analytics**
- **SQL & Pandas Practice**
- **GroupBy Operations**
- **Data Visualization**
- **KPI Analysis**
- **Dashboard Development**
- **Business Intelligence Projects**

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**Dataset Columns Description**

| Column Name | Description |
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| POS | Ranking position of player in season |
| player_name | Name of the batsman |
| team | IPL franchise/team |
| RUNS | Total runs scored |
| MAT | Matches played |
| INNS | Innings played |
| NO | Number of not outs |
| HS | Highest score |

| Column Name | Description |
|---|---|
| AVG | Batting average |
| BF | Balls faced |
| SR | Strike rate |
| 100 | Total centuries |
| 50 | Total half centuries |
| 4S | Total fours |
| 6S | Total sixes |
| season | IPL season year |
| status | Out / Not Out statu |

**CATEGORY 1 — Basic Data Exploration Questions**

1. Find the total number of players in the dataset.
2. Find the total number of teams.
3. Display all unique IPL seasons available in the dataset.
4. Find the total runs scored by all players.
5. Find the average strike rate of all players.
6. Find the player with the highest score.
7. Find the player with the lowest batting average.
8. Count total centuries scored in the dataset.
9. Count total half centuries scored in the dataset.
10. Find total fours and sixes hit in all seasons.

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**CATEGORY 2 — Team-wise Analysis Questions**

11. Find the total runs scored by each team.

12. Find the average strike rate of each team.
13. Find the team with the highest batting average.
14. Find the team that scored the most sixes.
15. Find the team that scored the most fours.
16. Find the total centuries scored by each team.
17. Find the total half centuries scored by each team.
18. Find the most aggressive team based on strike rate.
19. Find the team with the highest boundary percentage.
20. Find the team with the highest average runs per player.

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**CATEGORY 3 — Player-wise Analysis Questions**

21. Find the top 10 run scorers in IPL history.
22. Find players with the highest batting average.
23. Find players with the highest strike rate.
24. Find players with the most centuries.
25. Find players with the most half centuries.
26. Find players with the most sixes.
27. Find players with the most fours.
28. Find players with the highest individual score.
29. Find players who remained not out most times.
30. Find players who played the most matches.

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**CATEGORY 4 — Season-wise Analysis Questions**

31. Find total runs scored in each IPL season.
32. Find total sixes hit in each season.
33. Find total fours hit in each season.

34. Find the season with the highest batting average.
35. Find the season with the highest strike rate.
36. Find the highest run scorer in each season.
37. Find total centuries scored season-wise.
38. Find total half centuries scored season-wise.
39. Find the most aggressive batting season.
40. Find the season with the highest total boundaries.

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**CATEGORY 5 — Team + Season Combined Analysis**

41. Find total runs scored by each team in every season.
42. Find the best performing team in each season.
43. Find season-wise strike rate of every team.
44. Find season-wise boundary percentage of teams.
45. Find the team with highest runs in a single season.
46. Find the team with lowest runs in a single season.
47. Compare team batting averages across seasons.
48. Find season-wise sixes scored by each team.
49. Find season-wise fours scored by each team.
50. Find teams with consistent performance across seasons.

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**CATEGORY 6 — Advanced GroupBy Analysis Questions**

51. Find players with consistent performance using standard deviation.
52. Rank players within each team based on runs.
53. Find top 3 players from every team.
54. Find top scorer from each season.
55. Find players whose strike rate is above team average.

56. Find players whose runs are above season average.
57. Find players contributing highest percentage of team runs.
58. Find the player with maximum boundary dependency.
59. Find players with highest runs per match ratio.
60. Find players with highest sixes per innings ratio.

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**CATEGORY 7 — Pivot Table & Crosstab Questions**

61. Create a pivot table for team vs season total runs.
62. Create a pivot table for team vs season average strike rate.
63. Create a pivot table for player vs season runs.
64. Create a pivot table for team vs season sixes.
65. Create a pivot table for team vs season fours.
66. Create a crosstab between team and status.
67. Create a crosstab between season and centuries scored.
68. Create a pivot table for player vs team runs.
69. Create a pivot table for player vs season batting average.
70. Create a pivot table showing top run scorers season-wise.

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**CATEGORY 8 — Descriptive Statistical Analysis Questions**

71. Find mean, median, and mode of runs.
72. Find variance and standard deviation of strike rate.
73. Find quartiles of batting average.
74. Find distribution of runs scored by players.
75. Find skewness of strike rate.
76. Find correlation between runs and strike rate.
77. Find correlation between fours and sixes.

78. Find outliers in batting average.
79. Find outliers in strike rate.
80. Find players performing above the 75th percentile.

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**CATEGORY 9 — Data Cleaning & Preparation Questions**

81. Check for missing values in the dataset.
82. Replace null values with appropriate methods.
83. Convert numeric columns into proper datatype.
84. Remove duplicate records from dataset.
85. Rename all columns into lowercase format.
86. Create a new column called boundary_runs.
87. Create a new column called boundary_percentage.
88. Create a new column called runs_per_match.
89. Create a new column called sixes_per_innings.
90. Create a new column called batting_impact_score.

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**CATEGORY 10 — Visualization-Based Questions**

91. Create a bar chart of top 10 run scorers.
92. Create a pie chart of team-wise runs contribution.
93. Create a line chart of season-wise total runs.
94. Create a histogram of strike rate distribution.
95. Create a scatter plot between runs and strike rate.
96. Create a heatmap of team vs season runs.
97. Create a boxplot of batting average by teams.
98. Create a countplot of centuries by season.
99. Create a stacked bar chart for fours and sixes by team.

100. Create a dashboard summarizing IPL batting analysis.